

Name: _____

The Ripple Effect — The Work of Water Across New York State Surfaces

Student Answer Packet 2: To be completed individually.

You may use the information from **Student Directions** and **Student Answer Packet 1** to help you answer some of these questions.

QUESTIONS:

Note: You will use your OWN data to answer the question(s) associated with your Investigation Option. You will use the CLASS data for the remainder of the questions, if needed.

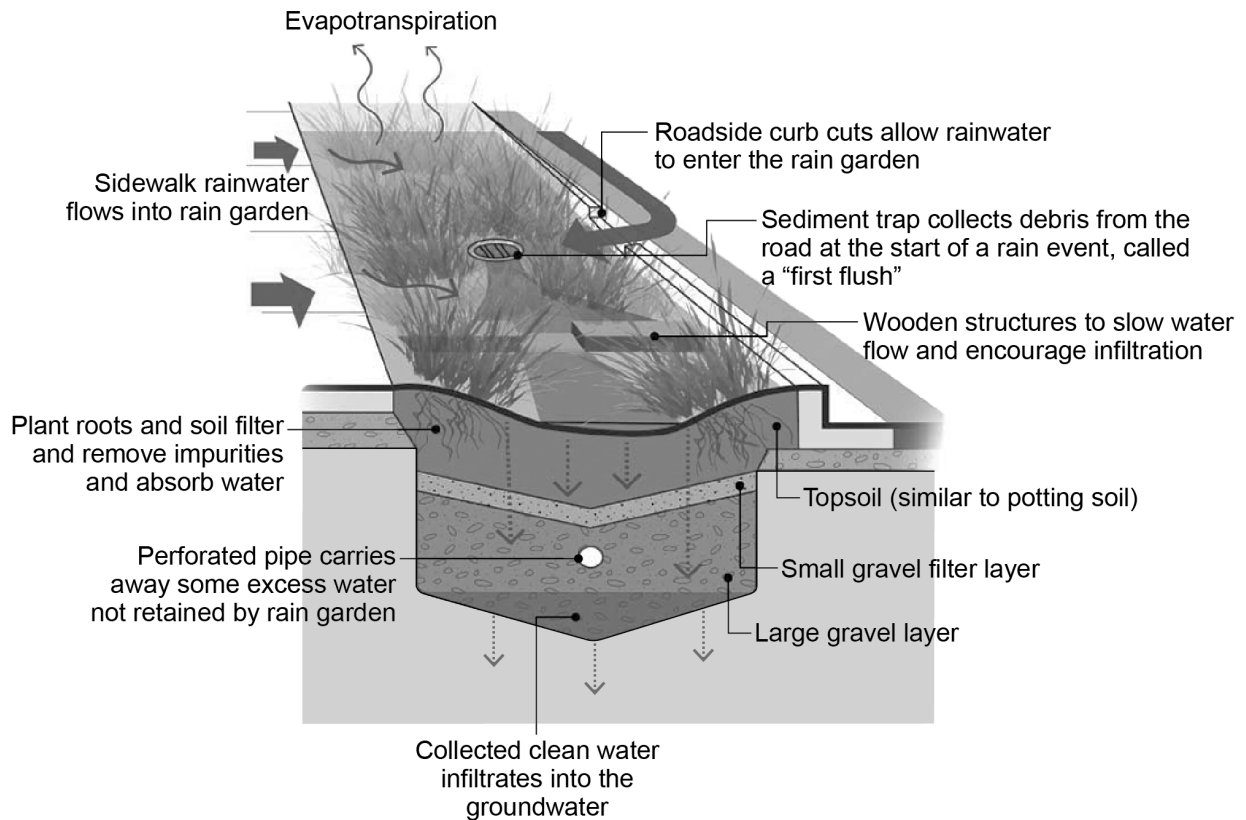
1. Using data from **Investigation Option A: Stream Table**, describe the general relationship between the independent variable and the dependent variable that was investigated.

2. Using the photograph of the Genesee River (found on page 3 of **Student Directions**), explain how Earth's surface is changed through **both** constructive and destructive processes of deposition and erosion of sediment and rock as a result of the properties of water.

3. Using data from **Investigation Option B: Rain Garden**, describe the general relationship between the independent and the dependent variable that was investigated.

The model below represents the parts of a rain garden like those found in New York City. Arrows represent the movement of water, in its different forms, through the system.

Green Infrastructure Rain Garden Model



Constructed Rain Garden



- 4a. Use information from your data table *or* the model to describe how the permeability of different materials used in a rain garden increase the amount of infiltration of water and decrease the amount of runoff.

- 4b. A student closed the bottom of their rain garden model so that **no** water drained out and the porosity of the rain garden could be determined. How could the rain garden materials be used to determine the porosity of the sediments used in the rain garden?

- 4c. How is porosity of Earth materials important in an effective rain garden design?

5. Other than an interaction between infiltration and runoff, how can rain gardens change **one** water cycle process, causing a feedback change in the geosphere of urban settings? Include **both** a water cycle process and a feedback change in your answer.

6. Using data from **Investigation C: Weathering of Earth Materials**, describe how the type of liquid affects the amount of weathering of the chalk pieces.

7. Other than the variables that could have been investigated in **Investigation C**, describe **one** unique physical *or* chemical property of water that contributes to the weathering of rocks. Then describe how this property changes the surface of **one** landscape region in New York State (found on page 4 of ***Student Directions***).

8. Explain how the effects of water within the Genesee River basin (found on page 4 of ***Student Directions***) occur at different spatial *and* temporal scales compared to the effects of water in the development of the many caves found in New York State's Allegheny Plateau region (found on page 11 of ***Student Directions***).

Genesee River

Spatial: _____

Temporal: _____

Allegheny Plateau Caves

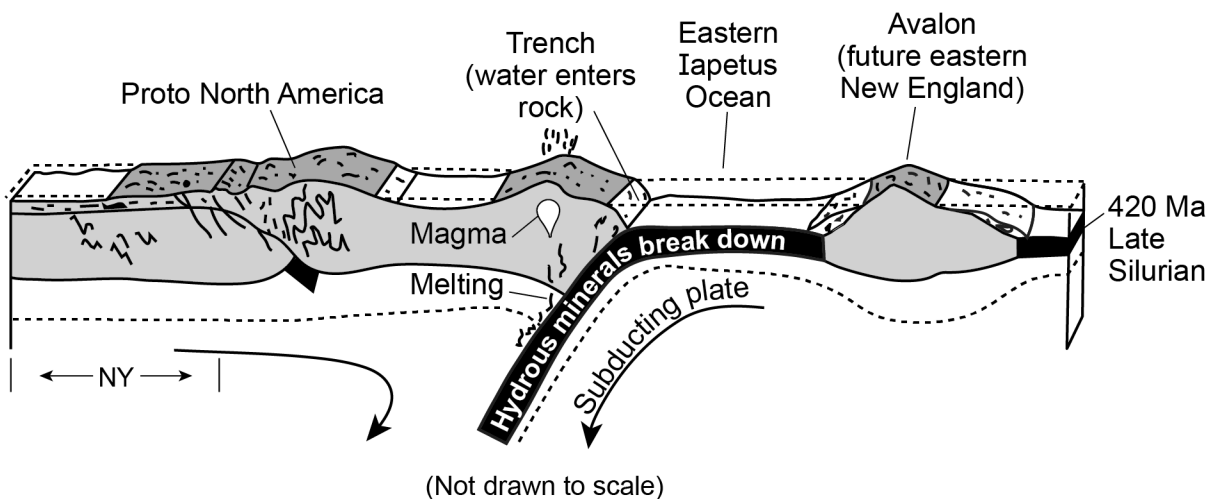
Spatial: _____

Temporal: _____

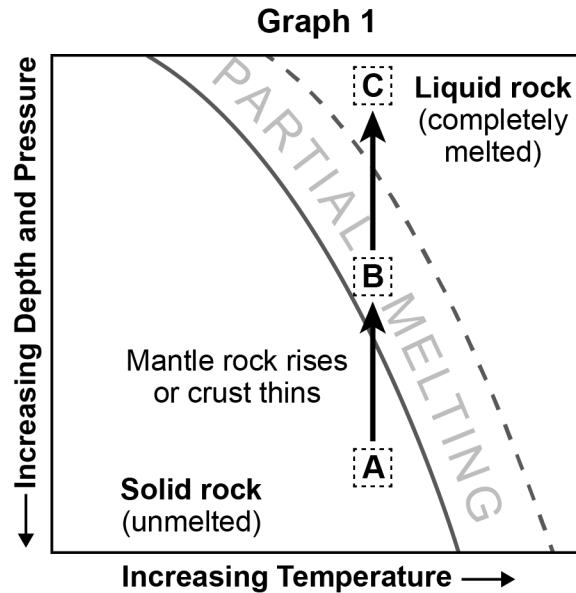
Before 450 million years ago, North America was missing most of what is now New England. Formed over millions of years, New England was slowly pieced together from strips of land added to the proto-North American continent. This land, called exotic terrane, originated from other places that were added to the continent as tectonic plates converged and the Iapetus Ocean closed. Avalon is one exotic terrane that is now part of eastern New England. New York's location on proto-North America approximately 450 million years ago is indicated.

Liquid water not only affects Earth's surface materials and processes but can also impact the melting point and viscosities of rocks below Earth's surface, specifically within the mantle. The water reacts with the rocks to form hydrous minerals (minerals that contain water). These minerals become unstable as pressure and temperature increase beneath the surface. The water is then released into the mantle rock above the slab, creating wet mantle rock conditions rising toward Earth's surface.

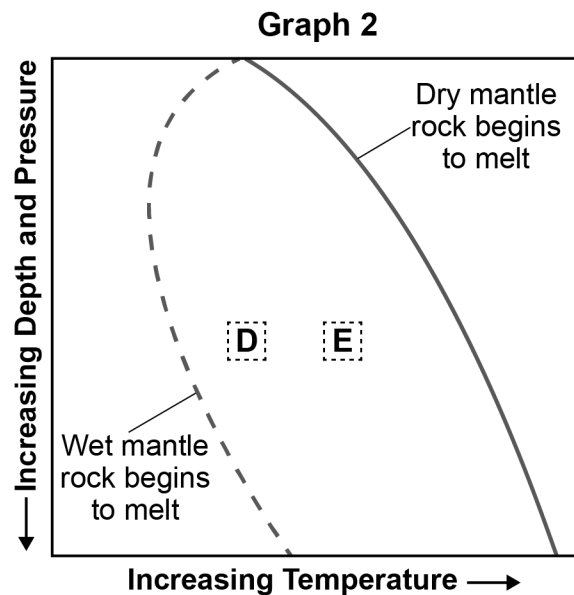
Formation of Hydrous Minerals at a Subduction Zone



Graph 1 shows the effect of decreasing temperature and pressure on mantle rock that does not contain water. Boxes *A*, *B*, and *C* show the progression of solid rock to liquid rock as rock rises closer to Earth's surface and depth and pressure decrease. The solid line represents the conditions necessary for dry solid mantle rock to begin to melt. The dashed line represents the conditions necessary for solid mantle rock to completely melt and become a liquid.



Graph 2 compares the temperature at which wet mantle rock and dry mantle rock begin to melt as depth and pressure increase. Boxes *D* and *E* represent mantle rock at the same depth and pressure conditions. Box *D* represents wet mantle rock and Box *E* represents dry mantle rock.



9. Using evidence from the graphs, determine the state of matter for mantle rocks *D* and *E*, and provide an explanation for how the presence of water in mantle rock affects the melting point.

Mantle Rock D: _____

Mantle Rock E: _____

Explanation: _____
